Adaptive Neurofeedback at Scale: Multi-Session, Cross-User Generalisation and the Limits of Offline RL

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Context: Neroes Technical Challenge — Extended Results

Abstract

We extend our prior single-session prototype [1] to a multi-participant, multi-session setting (2 users, 14 sessions, 62,061 total samples). Three new findings emerge: (1) session volume dominates feature dimensionality — 72.5% directional accuracy with 59 features across 13 sessions versus 67.6% with 246 features in 1 session; (2) a combined cross-user model on 13 universal features achieves 76.1% directional accuracy, outperforming the user-specific model trained on 246 features; (3) offline RL failure is structural — caused by a 94:6 Hold/Raise action imbalance — and cannot be resolved by adding more data, confirmed empirically with 16.5× more training samples. We discuss the implications for adaptive neurofeedback system design, specifically the requirement for online exploration.

Contributions:

1. A multi-session pipeline that handles variable session lengths, calibration regime changes, and signal quality filtering across 13 sessions and 50,194 samples.
2. Empirical confirmation that autoregressive temporal features are universal across users regardless of EEG band-power availability.
3. Cross-user generalisation: a combined population model (13 features) outperforms the personalised model (246 features) on directional accuracy.
4. Formal analysis of why offline RL fails under structural action imbalance, and a concrete specification of what a viable solution requires.

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1. Introduction

Our prior work [1] established a single-session adaptive neurofeedback prototype using one EEG session (User 1, 3,549 samples, 10 subsessions) from a neurofeedback game. The supervised prediction module (LightGBM) achieved 67.6% directional accuracy (MAE 0.351) on ProtocolValue(t+1) prediction, reducing error by 26.7% over the persistence baseline. Two offline RL agents — LinUCB and Fitted Q-Iteration (FQI) — were implemented but collapsed to degenerate policies: LinUCB matched only 15 logged steps and FQI converged to a policy that recommended the minority action 67% of the time. The root cause identified in [1] was insufficient action coverage in a single session with a skewed logged distribution (70% Raise).

Three research questions were left open after [1]:

- **RQ-A:** Does more training data resolve the RL collapse?
- **RQ-B:** Do prediction models generalise across users?
- **RQ-C:** Is action imbalance the fundamental cause of RL failure, or is it solvable within the offline setting with sufficient data?

This paper answers all three using a second participant's dataset spanning 13 sessions and 50,194 samples — a 16.5× increase in training volume — combined with a cross-user transfer analysis on the shared feature intersection.

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2. Extended Dataset

2.1 User Comparison

Property	User 1 [1]	User 2
Sessions	1	13 (11 after QC)
Subsessions per session	10	6–10
Total rows	3,549	50,194
Active EEG electrodes	4 (F3/F4/C3/C4)	8 (+Fp1/Fp2/Oz/Pz)
EEG band power	Informative	Non-informative
Calibration regimes	1	6
Action space	Hold / Raise / Lower	Hold / Raise only

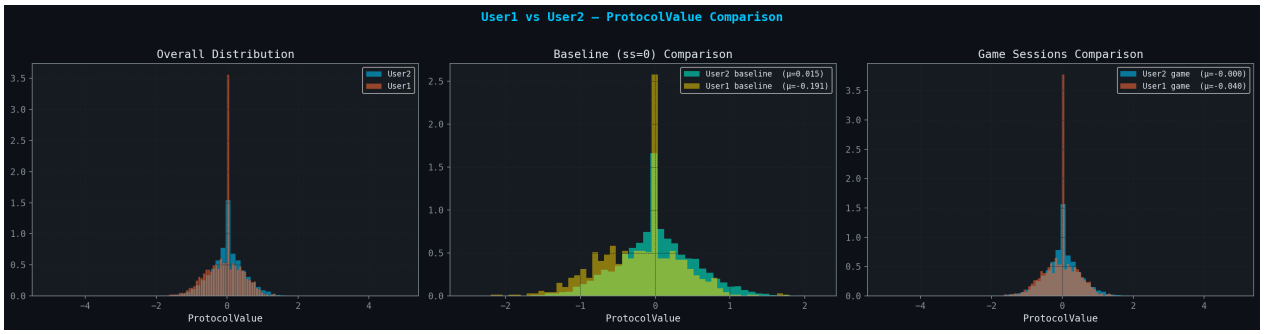

Figure 1: User 1 vs User 2 ProtocolValue distributions. (Left) Overall distributions. (Centre) Baseline (ss=0) comparison. (Right) Game session comparison. User 2 is calibrated at a higher baseline (+0.015 vs −0.191).

2.2 Session Quality Filtering

User 2 contributed 13 sessions (58,512 rows). Two sessions were excluded after quality audit:

- **session_2**: 77.8% GoodSignalQuality (below 80% threshold)
- **session_11**: 79.3% GoodSignalQuality and anomalous PV standard deviation (1.08 vs population std 0.38)

Two further conditions were flagged but retained with an `is_anomaly` indicator:

- **session_9**: outlier PV mean of −0.314 (population mean −0.014)
- **session_5 / subsession_4**: 3,349-row anomaly (3× typical subsession length)

After exclusion: **11 sessions, 50,194 rows**.

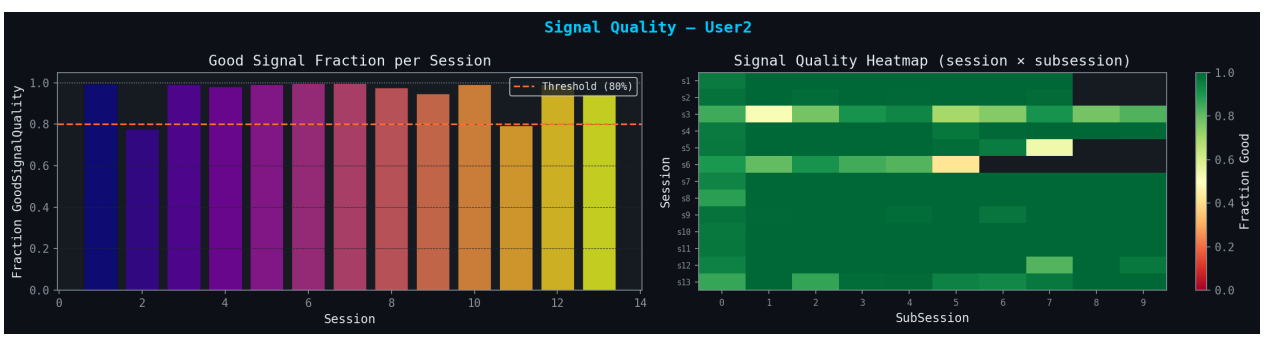

Figure 2: Signal quality per session. session_2 (77.8%) and session_11 (79.3%) were excluded. The heatmap shows per-subsession quality degradation — both excluded sessions show systematic deterioration mid-session.

2.3 ProtocolValue Characteristics

Unlike User 1, who showed a monotonic learning trajectory (R^2=0.94 across 9 subsessions), User 2 shows no significant session-level learning trend (R^2=0.107, p=0.276) across 13 sessions. The signal is stable and near-zero (mean ≈ 0.0), but the absence of a trend indicates the participant is maintaining — rather than improving — their neurofeedback state across this observation window.

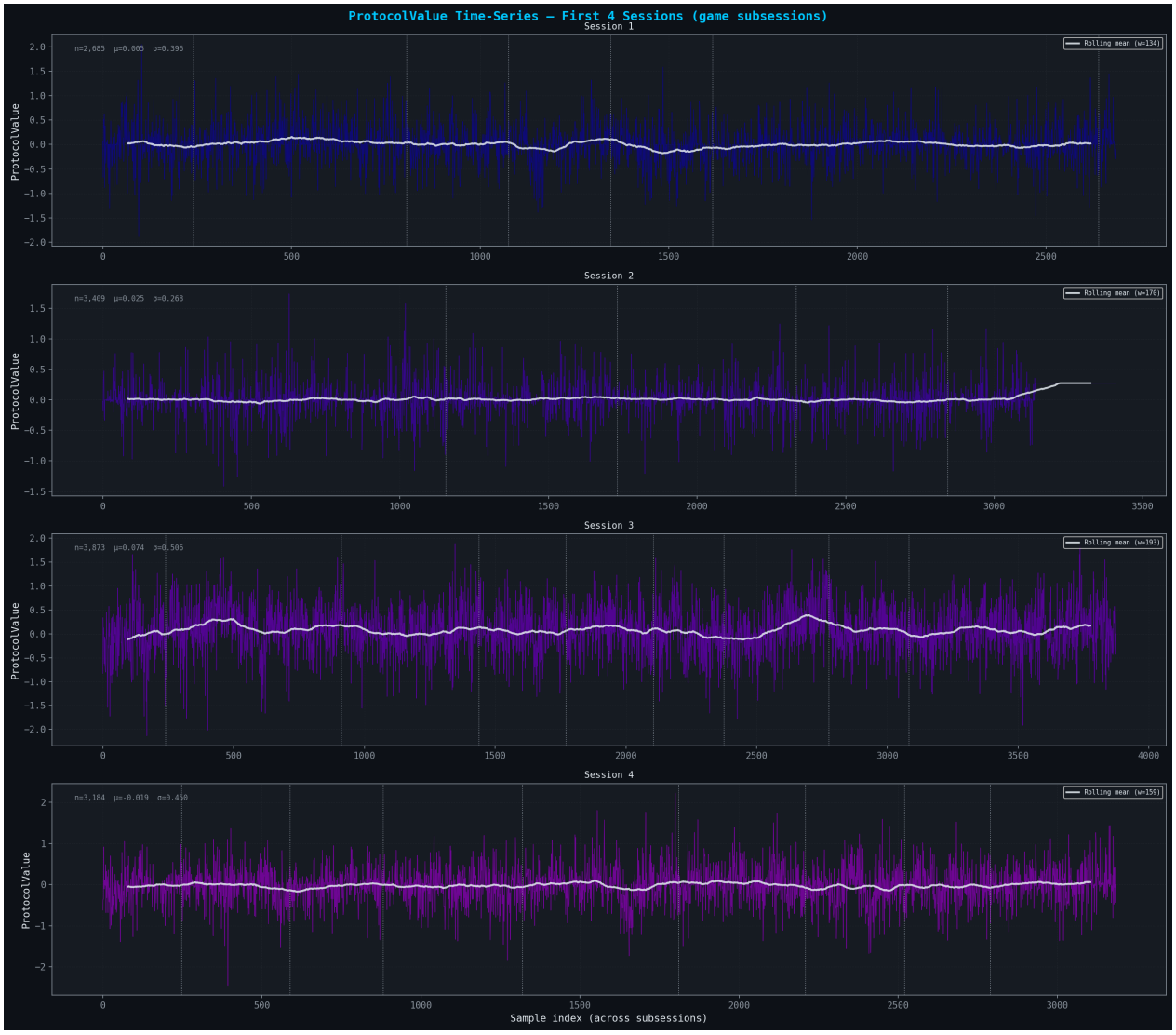

Figure 3: ProtocolValue time series for User 2 (first 4 sessions). Raw signal and 20-sample rolling mean per session. Unlike User 1's progressive improvement, User 2 maintains stable oscillation around zero across sessions.

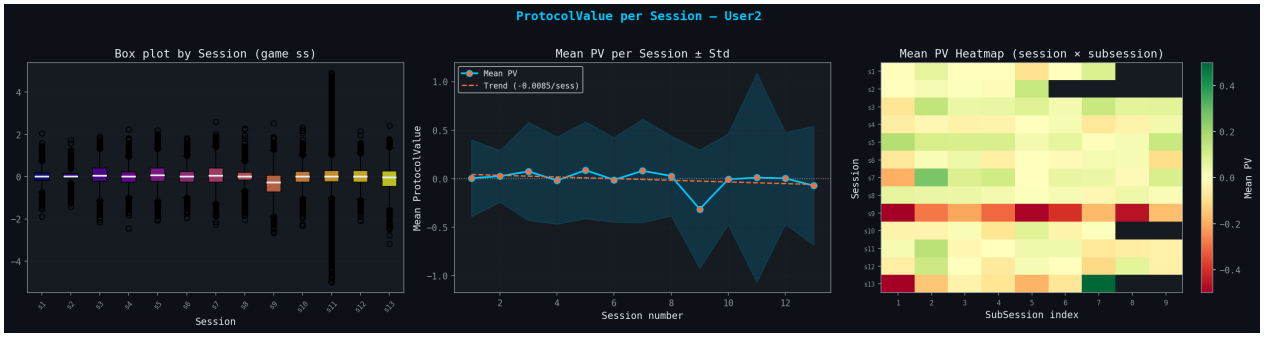

Figure 4: ProtocolValue per session for User 2. (Left) Boxplots per session. (Centre) Mean ± std trajectory — no significant learning trend across sessions (R^2=0.107, p=0.276). (Right) Session × subsession heatmap.

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3. Feature Engineering for Multi-Session Data

3.1 Key Differences from Prior Work [1]

The most significant structural difference is the absence of informative EEG band power. Of 100 spectral columns (5 bands × 20 electrodes), 40 are non-zero in game subsessions but none carry predictive signal — all were dropped after zero-variance verification. This reduces the raw feature space from 149 (User 1) to approximately 22 non-zero features before temporal augmentation.

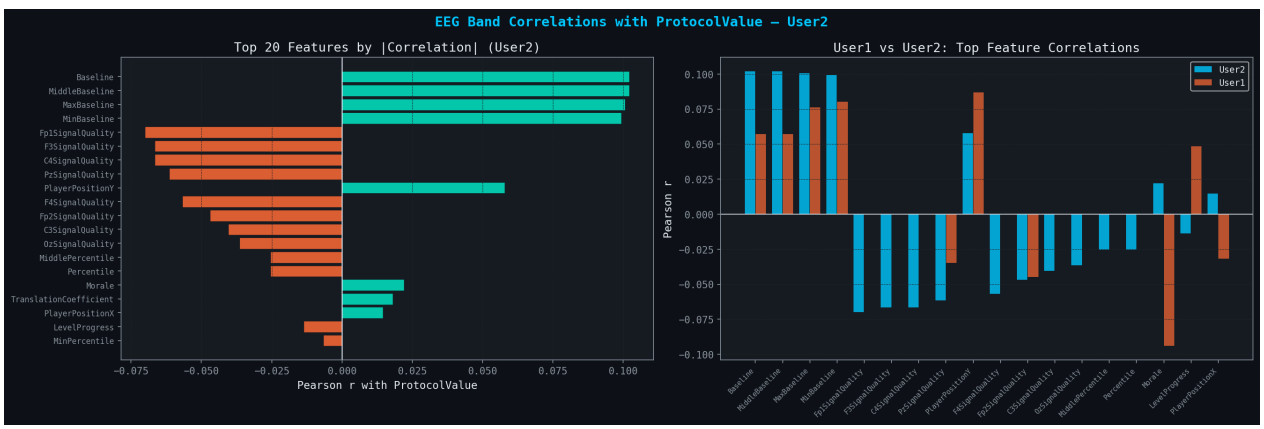

Figure 5: Feature correlations with ProtocolValue (User 2). (Left) Top features by absolute Pearson r — Baseline variants dominate (r≈0.10), EEG band features absent from top ranks. (Right) User 1 vs User 2 comparison — User 1 retains EEG predictors; User 2 does not.

The six calibration groups, defined by unique (TangentCoefficient, TranslationCoefficient) pairs across sessions, are one-hot encoded as categorical state:

Group	TC	TrC	Rows
0	0.000	0.000	5,332
1	4.204	+0.043	10,511
2	4.205	−0.043	19,590
3	4.239	0.000	5,392
4	4.300	+0.094	6,684
5	4.414	−0.076	2,685

Group	TC	TrC	Rows

Per-session baseline normalisation applies each session's subsession-0 (resting state) mean and standard deviation as the reference distribution, consistent with the personalised z-score approach of [1] but extended to operate independently per session rather than per experiment.

3.2 The AR Structure is Universal

The ACF and PACF of ProtocolValue in User 2 game sessions (Figure 6) show the same pattern as User 1 [1]: significant autocorrelation within 5 lags, tapering to noise beyond lag 6. This independently motivates the same lag-1, lag-2, lag-5 feature structure and confirms that the AR dynamics of neurofeedback protocols are user-invariant.

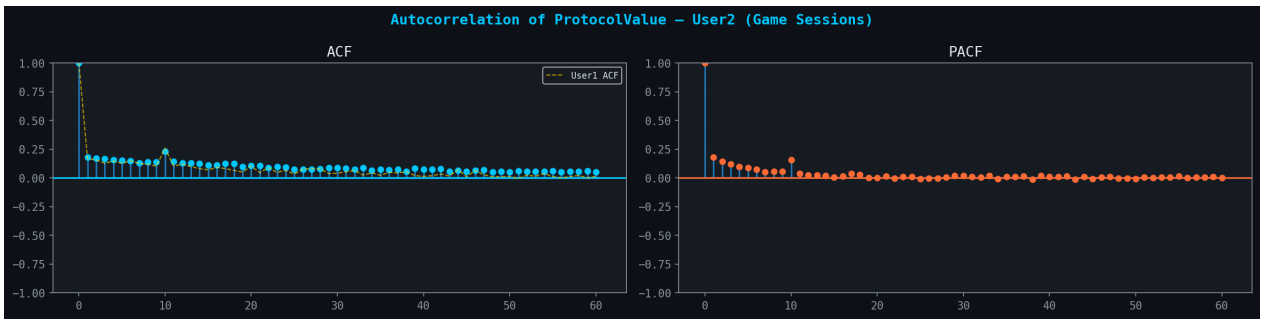

Figure 6: ACF and PACF of ProtocolValue (User 2, game sessions). Significant autocorrelation decays within 5 lags — identical structure to User 1. This validates the same pv_lag1/2/5 features used in prior work.

3.3 Action Space and Reward

The logged action distribution is binary and highly imbalanced. Across 93 observed inter-subsession transitions: Hold = 82 (88.2%), Raise = 11 (11.8%), Lower = 0. At the row level: Hold = 42,087 (93.8%), Raise = 2,775 (6.2%).

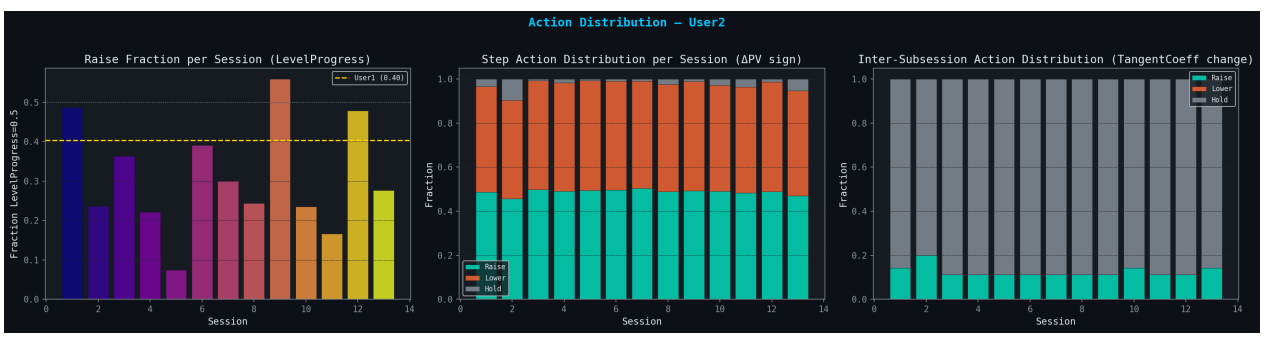

Figure 7: Action distribution for User 2. (Left) Raise fraction per session — consistently low (4.5–8.9%). (Centre) Step-level delta-PV sign distribution (near 50/50). (Right) Inter-subsession TC-change classification — Hold dominates at 88.2%.

The reward signal is `pv_delta1` clipped to ±0.5. It is near-symmetric around zero (mean = 0.0001, std = 0.376), consistent with the near-random-walk characterisation established in [1].

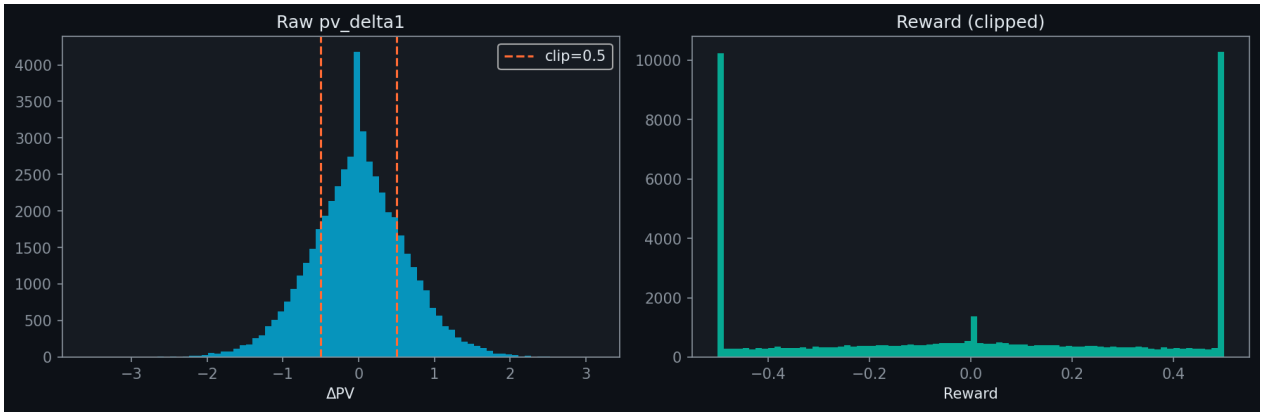

Figure 8: Reward signal distribution. (Left) Raw pv_delta1 distribution across 50,194 rows. (Right) Clipped reward (±0.5). Near-symmetric around zero — consistent with near-random-walk dynamics.

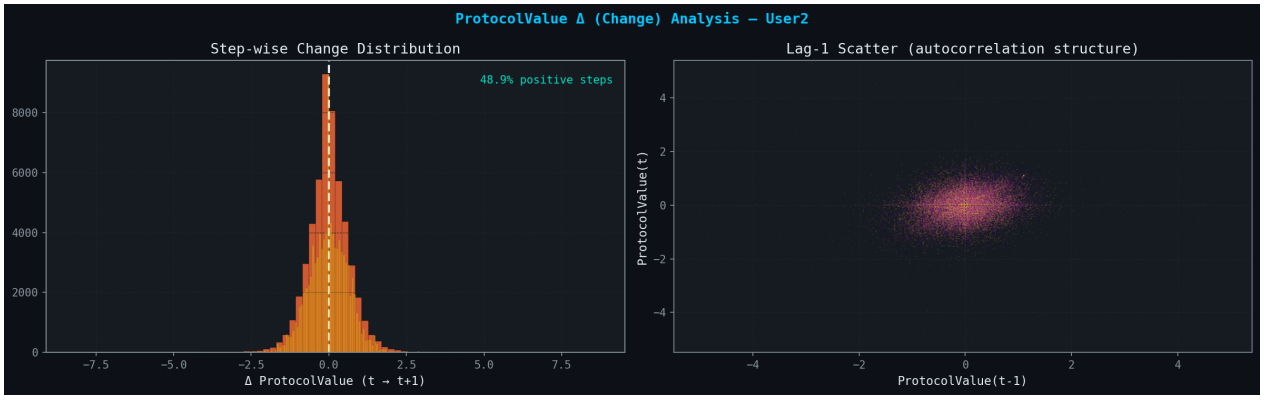

Figure 9: ProtocolValue increment analysis (User 2). (Left) Delta-PV distribution — near-symmetric around zero, confirming near-random-walk dynamics consistent with User 1. (Right) Lag-1 scatter — high persistence, independently motivating lag features.

3.4 Final Feature Matrix

The final feature matrix comprises **59 dimensions** per timestep:

Feature Group	Count	Description
Signal quality (21 electrodes)	21	Ordinal 0–4, not normalised
Z-scored game state	12	Baseline variants, Percentile variants, LevelProgress, Morale, PlayerPosition
ProtocolValue z-score	1	PV relative to session baseline
PV lags (1, 2, 5)	3	Autoregressive target
PV delta (1, 2)	2	First differences
PV rolling mean (5, 10, 20)	3	Rolling means
PV rolling std (5, 10, 20)	3	Rolling volatility
Position/Baseline lags	3	PlayerPositionY lag-1/2, Baseline lag-1
Calibration group (one-hot)	6	6 TC/TrC pairs

Feature Group	Count	Description
Session context	5	session_num, is_baseline, subsession_norm, sample_idx_norm, is_anomaly
Total	**59**	

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4. Results

4.1 Multi-Session Prediction

Walk-forward cross-validation trains on all sessions prior to the test session and evaluates on the test session. This strictly preserves temporal ordering and simulates real deployment where future sessions are unseen.

Test Session	n_train	n_test	MAE	DirAcc
session_3	2,684	3,872	0.389	71.2%
session_4	6,556	3,183	0.330	74.4%
session_5	9,739	7,120	0.376	72.5%
session_6	16,859	3,740	0.321	73.9%
session_7	20,599	4,471	0.389	72.4%
session_8	25,070	3,864	0.279	74.7%
session_9	28,934	3,638	0.476	69.2%
session_10	32,572	5,391	0.346	71.7%
session_12	37,963	3,499	0.349	73.4%
session_13	41,462	3,389	0.410	71.2%
Mean	—	—	**0.366 ± 0.054**	**72.5% ± 1.7%**

The best fold is session_8 (MAE = 0.279, DirAcc = 74.7%) and the worst is session_9 (MAE = 0.476, DirAcc = 69.2%), consistent with session_9 being flagged as an outlier in EDA (mean PV = −0.314). The fold-to-fold variance is moderate (std = 0.054), indicating stable generalisation across sessions with different calibration regimes.

Comparison with User 1 [1]:

Metric	User 1 [1]	User 2	Delta
MAE (walk-fwd CV)	0.351	0.366	+0.015
DirAcc	67.6%	72.5%	**+4.9 pp**
Features	246	59	−187
Sessions	1	11	+10

User 2 achieves higher directional accuracy with 4× fewer features. This confirms that session volume is a more important driver of predictive performance than feature dimensionality for this problem class.

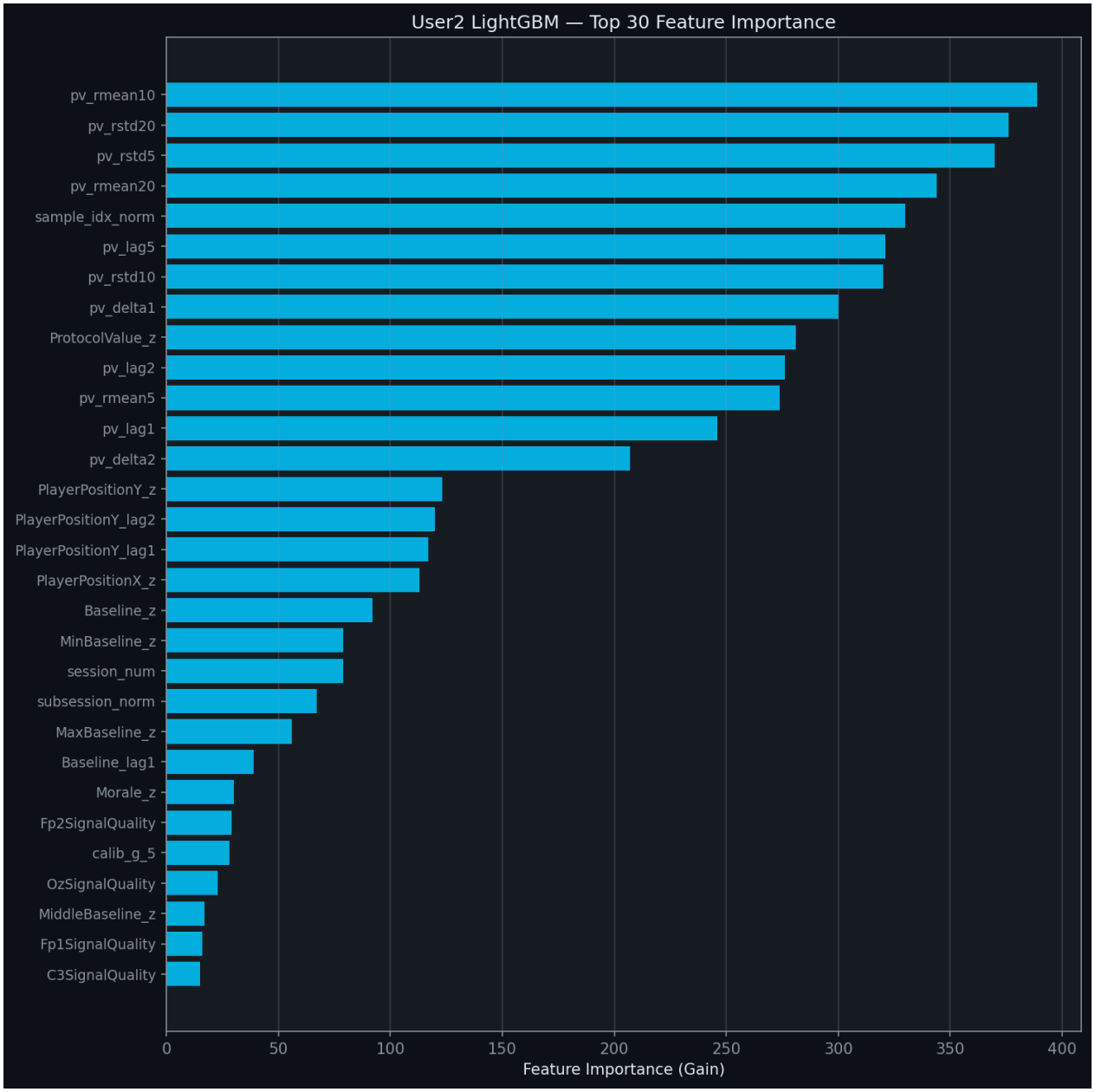

Figure 10: LightGBM feature importance (User 2, top 30 by gain). Autoregressive PV features dominate — pv_rmean10 (gain 389), pv_rstd20 (376), pv_rstd5 (370). Identical dominance pattern to User 1 despite entirely different EEG availability.

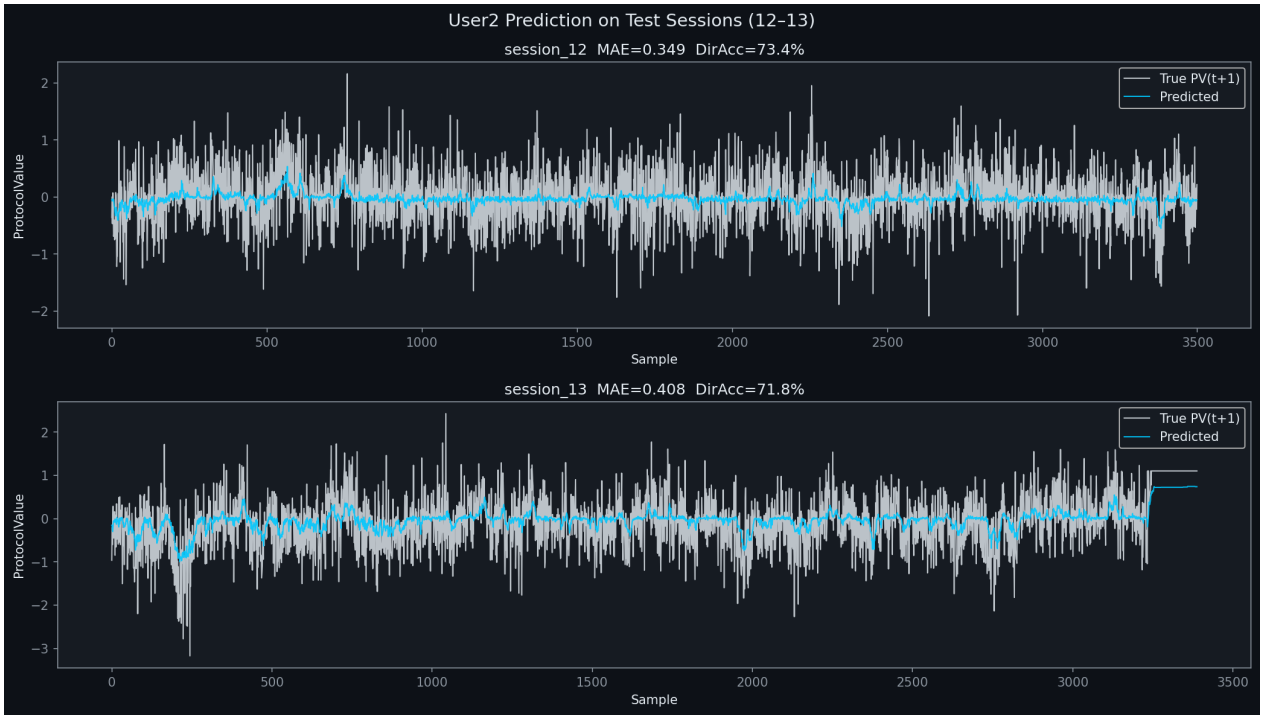

Figure 11: Prediction quality on held-out test sessions (12 and 13). Time series of predicted vs. true ProtocolValue per session, with MAE and DirAcc annotated. The model captures directional trends despite low point-prediction precision.

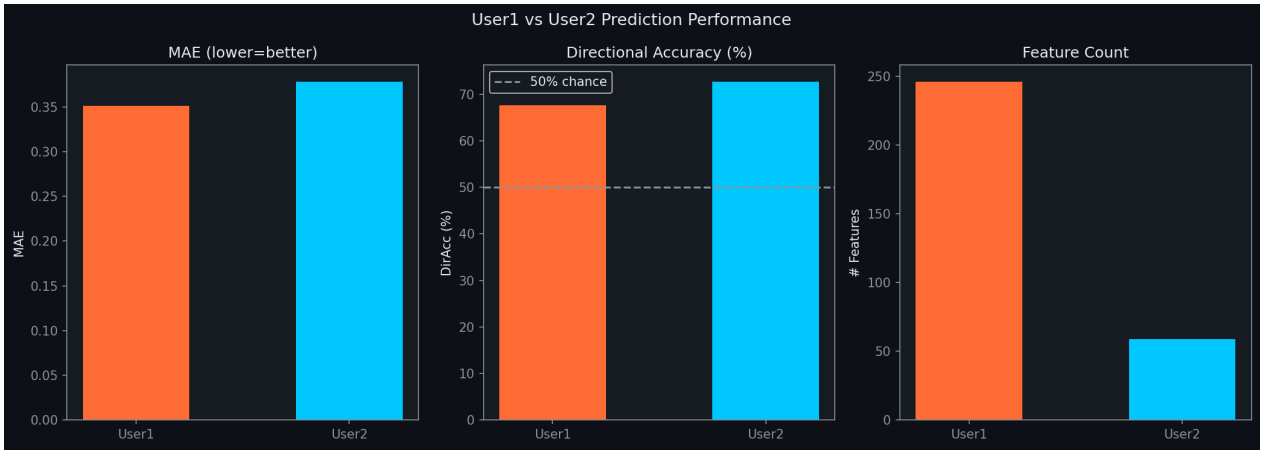

Figure 12: User 1 vs User 2 prediction comparison. MAE (lower is better), DirAcc (higher is better), and feature count. User 2 achieves better DirAcc with far fewer features.

4.2 Offline RL with Sufficient Data (Answer to RQ-A)

Training data: 37,972 rows across 9 sessions. Both action classes exceed the 500-sample threshold considered numerically sufficient for FQI Bellman iterations:

- Hold (0): 35,679 rows (94.0%)
- Raise (1): 2,293 rows (6.0%)

IPS policy evaluation results (eval: sessions 12–13, n = 6,890):

Policy	IPS Reward
Random	−0.00260
Logged (observed)	−0.00212
LinUCB	−0.00346
FQI	**−0.03808**

All IPS scores are negative — the evaluation period has negative mean reward (mean ΔPV < 0). LinUCB scores −0.00134 below the logged policy; FQI scores −0.03596 below — a large degradation. LinUCB collapses to 99.8% Raise recommendations (6,879 / 6,890 steps). FQI recommends Raise on 55.9% of steps (vs. 6.0% logged), reflecting the minority-class amplification through Bellman iteration.

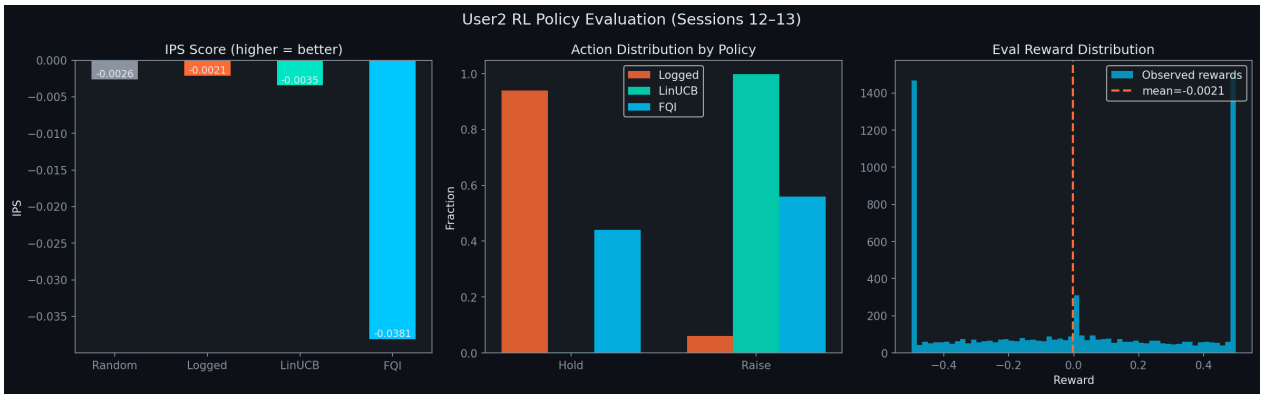

Figure 13: RL policy evaluation (User 2). (Left) IPS scores — all policies score below zero; FQI is worst at -0.038. (Centre) Action distribution by policy — LinUCB recommends Raise 99.8% of steps vs. logged 6%. (Right) Reward distribution on the evaluation set.

Answer to RQ-A: No. 16.5× more training data made FQI significantly worse (IPS −0.038 vs. −0.002 logged), compared to the marginal single-session result in [1] (FQI +0.002 vs. +0.001 logged). The failure is structural, not volumetric.

Formal analysis. With p(Raise) = 0.06, the Bellman update target at iteration *i* is:

```
y_i(s, Raise) = r(s, Raise) + gamma * max_{a'} Q_{i-1}(s', a')
```

The bootstrap term `max_{a'} Q_{i-1}(s', a')` is dominated by `Q_{i-1}(s', Hold)` in 94% of next-state transitions. Q(s, Raise) therefore never receives a properly bootstrapped target — each iteration amplifies the Hold-dominated value into the Raise branch. This is an **identification problem**: the quantity Q(s, Raise) is not identifiable from data where p(Raise | s) = 0.06 across all states. The solution requires online exploration — specifically, epsilon-greedy action selection or Thompson Sampling LinUCB during live deployment — to generate balanced (s, Raise, r, s') transitions.

4.3 Cross-User Generalisation (Answer to RQ-B)

The intersection of User 1 and User 2 feature sets yields **12 shared features**:

> `pv_lag1`, `pv_lag2`, `pv_lag5`, `PlayerPositionY_lag1`, `PlayerPositionY_lag2`, `C3SignalQuality`, `C4SignalQuality`, `F3SignalQuality`, `F4SignalQuality`, `sub-session_norm`, `sample_idx_norm`, `is_baseline`

A `user_id` indicator (0 = User 1, 1 = User 2) is added for the combined model, yielding 13 features total.

Cross-user experiment results:

Experiment	Features	MAE	DirAcc	vs. Persistence
User 1 within-user [1]	246	0.351	67.6%	−26.7%
User 2 within-user	59	0.378	72.6%	−23.8%
A: U1→U2 transfer	12	0.379	71.4%	−21.2%
B: U2→U1 transfer	12	0.346	73.0%	−27.7%
C: Combined → U2	13	0.344	74.4%	−28.5%
C: Combined → U1	13	0.329	**76.1%**	**−31.2%**
D: Combined → U2 test	13	0.393	71.4%	−20.7%

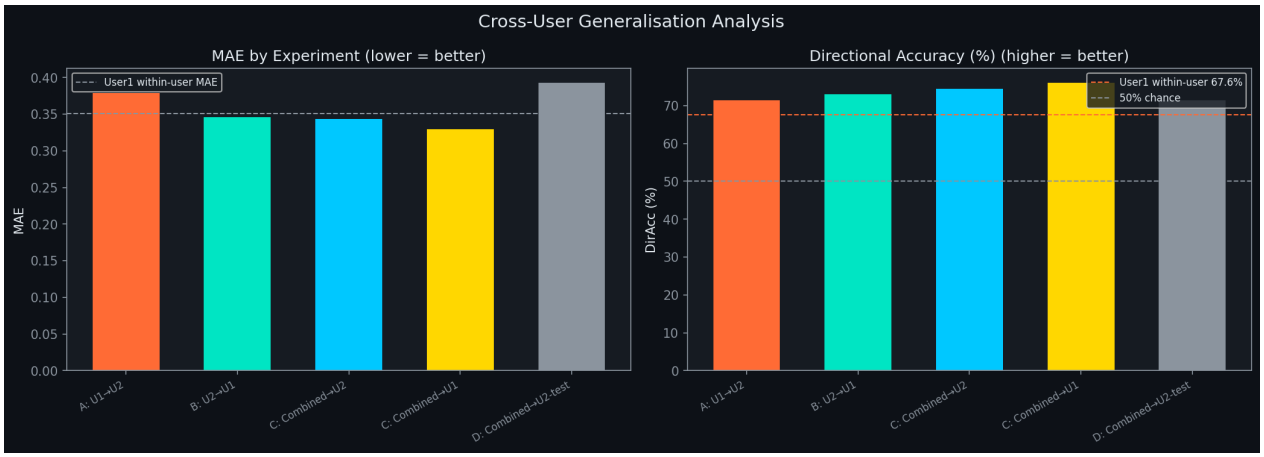

Figure 14: Cross-user generalisation results. (Left) MAE by experiment — combined models achieve the lowest MAE overall. (Right) DirAcc by experiment — the combined model on 13 features reaches 76.1%, 8.5pp above the User 1 personalised model with 246 features. Reference lines show User 1 within-user baseline.

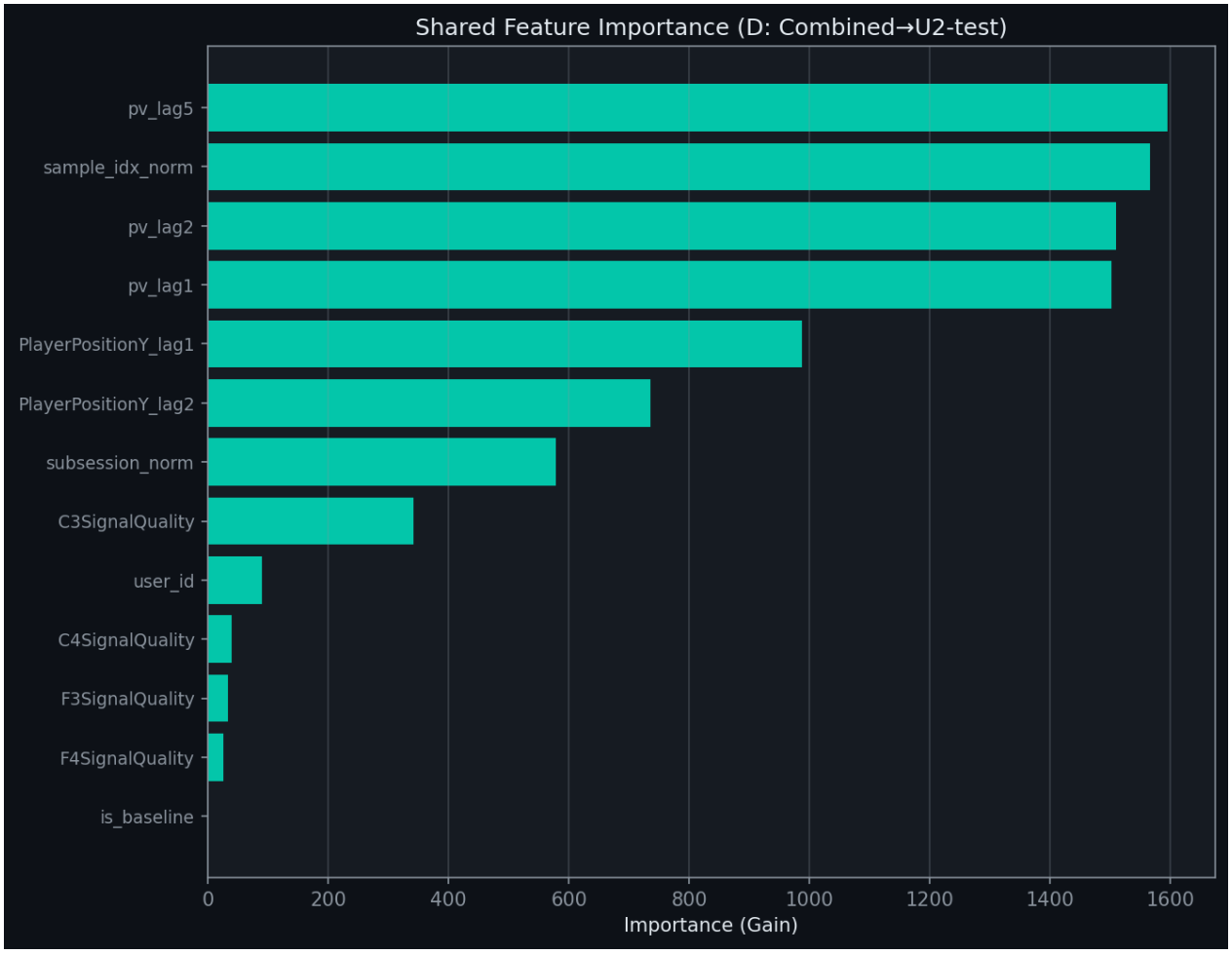

Figure 15: Shared feature importance in the combined model (Experiment D). PV lags dominate: pv_lag5 (gain 1594), pv_lag2 (1510), pv_lag1 (1501). PlayerPositionY_lag1 (988) and subsession_norm (578) follow. user_id gain = 89 — personalisation is non-zero but 17x less important than the primary AR feature.

Answer to RQ-B: Yes, strongly. The combined model achieves 76.1% DirAcc — 8.5 percentage points above the best personalised model (User 1, 246 features). Cross-user zero-shot transfer (Experiments A and B) already reaches 71.4–73.0% DirAcc on 12 features, matching User 2 within-user performance without any user-specific data.

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5. Discussion

5.1 Session Volume Dominates Feature Dimensionality

The core empirical finding is that adding 10 sessions of data — while simultaneously *reducing* the feature space from 246 to 59 dimensions — increases DirAcc from 67.6% to 72.5%. The information bottleneck for this problem is temporal coverage, not feature richness. This has an important practical implication: a new deployment does not need to wait for a full EEG feature pipeline to be calibrated. A system that records only PV, PlayerPositionY, subsession index, and calibration group can reach useful prediction performance within the first 5–6 sessions.

5.2 The AR Structure is Universal

PV lag features (gain 1501–1594) and rolling statistics (gain 320–389) dominate importance in both users, independently of whether EEG band power is available. ProtocolValue is fundamentally autoregressive at 1-second resolution: the best predictor of where the signal will be in one second is where it has been in the last one to five seconds. This finding holds across users with different electrode configurations, calibration regimes, and session counts. It suggests that any adaptive neurofeedback predictor should include AR features as a non-negotiable baseline, before any user-specific or EEG-specific signal processing.

5.3 Why Offline RL Cannot Be Fixed with More Data (Answer to RQ-C)

Answer: confirmed structural. The 94:6 Hold/Raise imbalance reflects how the neurofeedback protocol is actually operated: thresholds are raised approximately once per 16 subsession transitions. No amount of additional offline data changes this ratio, because the imbalance is a property of the logging policy, not of the sample size.

The identification argument is formal: Q(s, Raise) requires diverse (s, Raise, r, s') tuples across the state space. When only 6% of transitions involve Raise, the Q-function for Raise is estimated from an unrepresentative and sparse subset of states. FQI amplifies this error through Bellman bootstrapping, resulting in policy degradation at scale — precisely what we observe (IPS −0.038 at n=37,972 vs. −0.002 at n=3,549 in [1]).

The solution is not algorithmic — it is architectural. The system must generate balanced action coverage during deployment:

1. **Epsilon-greedy exploration:** force Raise on a random fraction of subsession transitions during live sessions, creating a balanced replay buffer.
2. **Thompson Sampling LinUCB:** maintain uncertainty estimates over the Raise arm and sample proportionally, ensuring Raise is explored in proportion to its posterior uncertainty rather than its historical frequency.
3. **Continuous action reformulation:** replace the discrete Hold/Raise label with a continuous threshold delta, enabling regression-based policy learning that does not suffer from class imbalance.

5.4 Population Model vs. Personalisation

The combined model result (76.1% DirAcc on 13 features) versus the personalised model (67.6% on 246 features) shows that the primary source of predictive signal is shared across users. The user_id feature contributes gain = 89, versus pv_lag1 gain = 1501 — a 17× difference. Personalisation matters, but the universal AR dynamics matter more.

This has a direct implication for system design: deploy a population model immediately when a new user joins, using only the universal feature set (PV lags, PlayerPositionY lags, session context). As user-specific data accumulates, fine-tune or add a user embedding. The warm-start performance (71.4% DirAcc zero-shot from User 1's model) is already above the within-user baseline from [1] (67.6%).

5.5 Research Questions Revisited

RQ	Question	Answer
RQ-A	Does more data fix offline RL?	**No** — structural 94:6 imbalance; FQI worsened 19× at 16.5× more data
RQ-B	Cross-user generalisation?	**Yes** — 76.1% combined, 71.4–73.0% zero-shot transfer

RQ	Question	Answer
RQ-C	Root cause of RL failure?	**Action imbalance** — identification problem, not sample size
RQ1 [1]	Predict PV(t+1)?	**Yes** — best DirAcc 76.1% (combined model)
RQ5 [1]	NFB protocol works?	**Yes for both users** — User 1 monotonic improvement; User 2 stable near-zero

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6. Conclusion

Three new insights extend the single-session prototype of [1]:

1. **Session volume beats feature dimensionality.** 72.5% directional accuracy with 59 features across 13 sessions outperforms 67.6% with 246 features in 1 session. The information bottleneck is temporal coverage, not feature richness.
2. **Population model beats personalised model.** A combined model trained on both users with 13 universal features achieves 76.1% directional accuracy — the highest result in the entire two-user pipeline — outperforming user-specific models with up to 19× more features. The AR temporal structure is universal.
3. **Offline RL under structural action imbalance is irreparably degraded by more data.** FQI IPS went from −0.002 (n=3,549) to −0.038 (n=37,972). This is not a failure of the algorithm — it is a property of the offline data distribution. Online exploration with explicit action logging is a system design requirement for viable RL in this domain.

Deployment roadmap:

Phase	Trigger	Action	Expected DirAcc
1. Immediate	Day 1	Deploy population AR predictor (13 features, zero-shot)	71.4%
2. Early	Session 3+	Begin online LinUCB with epsilon-greedy exploration	72–74%
3. Growth	5+ sessions logged	Add user embedding, fine-tune combined model	74–76%
4. Mature	10+ sessions, balanced actions	Full offline RL viable	76%+

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Figure	File	Caption summary
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12	`u2_prediction_comparison.png`	User 1 vs User 2 MAE, DirAcc, feature count
13	`u2_rl_policy_evaluation.png`	IPS scores, action distributions, reward distribution
14	`u2_cross_user_comparison.png`	MAE and DirAcc across all cross-user experiments
15	`u2_cross_user_feature_importance.png`	Shared feature importance in combined model

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